

# Wigger River Investigation

TIE

Research-ready investigation record for an uncertain freshwater eDNA detection.

|                        |                                |                                   |                            |
|------------------------|--------------------------------|-----------------------------------|----------------------------|
| Report status<br>Draft | Observation date<br>2014-06-25 | Prepared for<br>Scientific review | Scientific decision<br>TIE |
|------------------------|--------------------------------|-----------------------------------|----------------------------|

## Investigation Summary

A structured record for evaluating an uncertain eDNA detection and identifying a defensible follow-up action.

## Original eDNA Detection

| FIELD                        | VALUE                                   |
|------------------------------|-----------------------------------------|
| Species                      | Fredericella sultana                    |
| Station                      | S1                                      |
| Observation index            | 4                                       |
| Detection site reach         | 20446064                                |
| Coordinates                  | 47.31400039098192, 7.895400745863826    |
| Sampling date                | 2014-06-25                              |
| Historical state             | DETECTED                                |
| Concentration                | 1.2983219767633366e-17 mol/L            |
| Replicate and assay metadata | Unavailable in the verified H001 record |

## Evidence Assessment

| ZONE | EVIDENCE                           | SOURCE                                   | RULE                                 | RESULT   | EXPLANATION                                                                                                        | QUALITY         |
|------|------------------------------------|------------------------------------------|--------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------|-----------------|
| Z1   | historical_edna_measurement        | data_preflight/raw/carraro/eDNA_data.mat | No applicable rule                   | UNKNOWN  | No validated scientific rule applies to evidence type 'historical_edna_measurement'.                               | SOURCE_VERIFIED |
| Z1   | directed_hydrological_connectivity | HydroRIVERS NEXT_DOWN traversal          | hydrorivers.directed_contribution.v1 | SUPPORTS | Zone Z1 has a directed HydroRIVERS route to sampling reach 20446064; this supports topological compatibility only. | VERIFIED        |
| Z1   | directed_hydrological_connectivity | HydroRIVERS NEXT_DOWN traversal          | hydrorivers.directed_contribution.v1 | NEUTRAL  | Connectivity evidence targets zone root 20450127, not 20447392; it is neutral for zone Z1.                         | VERIFIED        |
| Z1   | directed_hydrological_connectivity | HydroRIVERS NEXT_DOWN traversal          | hydrorivers.directed_contribution.v1 | NEUTRAL  | Connectivity evidence targets zone root 20451169, not 20447392; it is neutral for zone Z1.                         | VERIFIED        |

| ZONE | EVIDENCE                           | SOURCE                                   | RULE                                 | RESULT   | EXPLANATION                                                                                                        | QUALITY         |
|------|------------------------------------|------------------------------------------|--------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------|-----------------|
| Z2   | historical_edna_measurement        | data_preflight/raw/carraro/eDNA_data.mat | No applicable rule                   | UNKNOWN  | No validated scientific rule applies to evidence type 'historical_edna_measurement'.                               | SOURCE_VERIFIED |
| Z2   | directed_hydrological_connectivity | HydroRIVERS NEXT_DOWN traversal          | hydrorivers.directed_contribution.v1 | NEUTRAL  | Connectivity evidence targets zone root 20447392, not 20450127; it is neutral for zone Z2.                         | VERIFIED        |
| Z2   | directed_hydrological_connectivity | HydroRIVERS NEXT_DOWN traversal          | hydrorivers.directed_contribution.v1 | SUPPORTS | Zone Z2 has a directed HydroRIVERS route to sampling reach 20446064; this supports topological compatibility only. | VERIFIED        |
| Z2   | directed_hydrological_connectivity | HydroRIVERS NEXT_DOWN traversal          | hydrorivers.directed_contribution.v1 | NEUTRAL  | Connectivity evidence targets zone root 20450127, not 20450127; it is neutral for zone Z2.                         | VERIFIED        |
| Z3   | historical_edna_measurement        | data_preflight/raw/carraro/eDNA_data.mat | No applicable rule                   | UNKNOWN  | No validated scientific rule applies to evidence type 'historical_edna_measurement'.                               | SOURCE_VERIFIED |
| Z3   | directed_hydrological_connectivity | HydroRIVERS NEXT_DOWN traversal          | hydrorivers.directed_contribution.v1 | NEUTRAL  | Connectivity evidence targets zone root 20447392, not 20451169; it is neutral for zone Z3.                         | VERIFIED        |
| Z3   | directed_hydrological_connectivity | HydroRIVERS NEXT_DOWN traversal          | hydrorivers.directed_contribution.v1 | NEUTRAL  | Connectivity evidence targets zone root 20450127, not 20451169; it is neutral for zone Z3.                         | VERIFIED        |
| Z3   | directed_hydrological_connectivity | HydroRIVERS NEXT_DOWN traversal          | hydrorivers.directed_contribution.v1 | SUPPORTS | Zone Z3 has a directed HydroRIVERS route to sampling reach 20446064; this supports topological compatibility only. | VERIFIED        |

## Hydrological Investigation

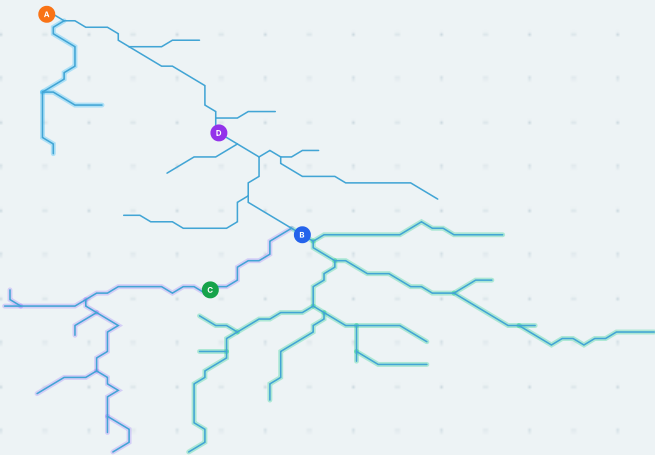
3 candidate zones and 4 sites are registered with validated reach identifiers. The validated map contains 49 reaches totaling 145.23 km.

Directed connectivity establishes a possible mapped transport pathway. It does not confirm the biological source.

## Investigation Map

### Verified Wigger River network

Verified preflight geometry · EPSG:4326



## Competing Source Explanations

| ZONE | ROOT REACH | VALIDATION |
|------|------------|------------|
| Z1   | 20447392   | VERIFIED   |
| Z2   | 20450127   | VERIFIED   |
| Z3   | 20451169   | VERIFIED   |

## Candidate Sampling Comparison

| REACH    | SIGNATURE | DISTINGUISHED PAIRS | PAIR SEPARATION | VALIDATION |
|----------|-----------|---------------------|-----------------|------------|
| 20451169 | [0, 0, 1] | Z1 vs Z3; Z2 vs Z3  | 2               | VERIFIED   |
| 20450127 | [0, 1, 0] | Z1 vs Z2; Z2 vs Z3  | 2               | VERIFIED   |
| 20446568 | [0, 1, 1] | Z1 vs Z2; Z1 vs Z3  | 2               | VERIFIED   |
| 20447392 | [1, 0, 0] | Z1 vs Z2; Z1 vs Z3  | 2               | VERIFIED   |

## Sampling Decision and Scientific Reasoning

### TIE

Multiple sites share the maximum topology-only pair separation score (2 pairs): Site B, Site C, Site D

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## Follow-up Sampling

Eligible tied sites under the persisted decision: Site B, Site C, Site D. Further sampling can test the remaining source hypotheses, but a negative sample does not prove absence and a positive sample does not by itself prove the exact source.

Field protocol, replicate count, assay, controls, collection time, and provenance must be supplied from the actual sampling design.

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## Environmental and One Health Context

Context does not independently establish source presence, pollution, disease, or human-health risk.

No environmental context records were collected for this report.

| STATUS   | SUPPORTED MONITORING FINDING                                                | POSSIBLE MONITORING ACTION                                                                        |
|----------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| OBSERVED | Fredericella sultana eDNA was detected in Carraro H001 at S1 on 2014-06-25. | F. sultana evidence establishes potential relevance for targeted parasite/fish-health monitoring. |

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## Data Sources

| SOURCE                                 | AVAILABILITY                             |
|----------------------------------------|------------------------------------------|
| Backend investigation records          | Loaded                                   |
| Verified Wigger preflight spatial data | Loaded with EPSG:4326 provenance         |
| Carraro historical eDNA source         | data_preflight/raw/carraro/eDNA_data.mat |
| Scientific rules                       | sampling.topology_pair_separation.v1     |

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## Assumptions

The supplied zones are the remaining hypotheses. Each hypothesis is represented by its validated root reach. A site outcome is binary directed reachability at HydroRIVERS reach resolution.

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## Limitations

Reachability is a mapped-network relation, not eDNA detection probability. The criterion compares binary outcomes and does not model transport, decay, abundance, field access, or analytical error.